

Theory Behind Simulations

Single VO2 Neuristor under Voltage and Current Sources

by Gabriel Berezovsky

Contents

1	Introduction	1
1.1	Current Dynamics Equation	1
1.2	Thermal Dynamics Equation	2
1.3	Resistance / Hysteresis Model	3
2	Derivation of the Current-Source Model	5
2.1	Starting Point: Original Voltage-Driven Equations	5
2.2	Assumptions and Circuit Interpretation	5
2.3	Step-by-Step Derivation for the Direct Current-Source Experiment	6
2.4	Final Continuous-Time Current-Driven Model	6
3	Discretization	7
3.1	Discretization of Current Dynamics Equation	7
3.2	Discretization of Thermal Dynamics Equation	8
3.3	Discretization of Resistance / Hysteresis Model	8
3.4	Discretization of the Current-Source Model	9
4	Fitting of the Resistance Model Based on Real Data	9
4.1	Input Data and Preprocessing	9
4.2	What We Obtain at Each Step vs What Must Be Assumed	10
4.3	Parameter Seeding and Intermediate Quantities	10
4.4	Optimization Objective and Fitting Procedure	11
4.5	Final Fitted Parameters, Quality Metrics, and Limits	13
5	Quasistatic and Dynamic-Phase Current-Source Models	14
5.1	Shared electrothermal equations	14
5.2	Original quasistatic phase response	14
5.3	Dynamic-phase working extension	15
6	Dynamic-Model Article-Comparison Campaign	15
6.1	Target observables	15
6.2	Search procedure	16
6.3	Timestep-convergence audit and numerical correction	17
6.4	Post-correction convergence results	18
6.5	Current scientific conclusion	19
7	Sources	20

1 Introduction

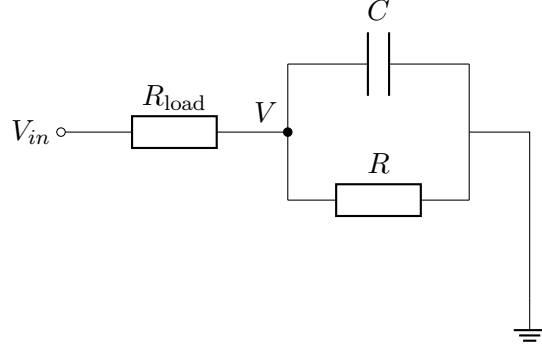


Figure 1: Single-neuristor equivalent circuit used in the voltage-driven model.

1.1 Current Dynamics Equation

Equation(s)

Source equation (Zhang *et al.*, 2024):

$$C \frac{dV_i}{dt} = \frac{V_i^{in}}{R_i^{load}} - V_i \left(\frac{1}{R_i} + \frac{1}{R_i^{load}} \right). \quad (1)$$

Single-device form used in this project:

$$C \frac{dV}{dt} = \frac{V_{in}}{R_{series}} - V \left(\frac{1}{R_{vo2}} + \frac{1}{R_{series}} \right), \quad R_{vo2} = R(T, \mathcal{H}), \quad (2)$$

$$\frac{dV}{dt} = \frac{V_{in} - V}{R_{series}C} - \frac{V}{R_{vo2}C}. \quad (3)$$

Variables and Units

Symbol	Meaning	Units
V_i, V	node/device voltage across VO ₂ and capacitor	V
V_i^{in}, V_{in}	applied source voltage	V
R_i, R_{vo2}	VO ₂ resistance state	Ω
R_i^{load}, R_{series}	series/load resistance	Ω
C	parasitic/effective node capacitance	F
t	time	s
$\mathcal{H}$	hysteresis internal state (branch/reversal memory)	–

Physical Interpretation

- $C dV/dt$: capacitive storage term (how fast node voltage can change).

- Source-branch current is $I_{src} = (V_{in} - V)/R_{series}$, which can be decomposed as $V_{in}/R_{series} - V/R_{series}$.
- V/R_{series} : the part of the source-branch term that opposes charging as node voltage rises.
- V/R_{vo2} : current through the VO₂ device; this term is strongly nonlinear because R_{vo2} depends on temperature and hysteresis state.

The competition between $I_{src} = (V_{in} - V)/R_{series}$ and VO₂ discharge determines oscillatory charging-switching dynamics.

1.2 Thermal Dynamics Equation

Equation(s)

Source equation (Zhang *et al.*, 2024):

$$C_{th} \frac{dT_i}{dt} = \frac{V_i^2}{R_i} - S_e(T_i - T_0) + S_c \nabla^2 T_i + \sigma \eta_i(t). \quad (4)$$

Single-device reduction ($\nabla^2 T = 0$):

$$C_{th} \frac{dT}{dt} = \frac{V^2}{R_{vo2}} - S_e(T - T_0) + \sigma \eta(t). \quad (5)$$

Variables and Units

Symbol	Meaning	Units
T_i, T	device temperature	K
t	time	s
C_{th}	thermal capacitance	J/K
V_i, V	device voltage	V
R_i, R_{vo2}	VO ₂ resistance	Ω
S_e	thermal conductance to environment/substrate	W/K
S_c	nearest-neighbor thermal coupling conductance	W/K
T_0	ambient/base temperature	K
$\nabla^2 T$	discrete thermal Laplacian (neighbor exchange term)	K
$\eta_i(t)$	Gaussian white-noise process (paper notation)	$s^{-1/2}$
σ	noise strength multiplier (paper notation)	$J s^{-1/2}$

Physical Interpretation

- $\frac{V^2}{R_{vo2}}$: Joule-heating input that increases T .
- $S_e(T - T_0)$: heat leakage to ambient that decreases T when $T > T_0$.

- $S_c \nabla^2 T$: diffusive thermal exchange with neighbors; positive when neighbors are hotter.
- $\sigma \eta_i(t)$: stochastic thermal fluctuations (paper noise term).

The thermal equation therefore balances Joule heating, environmental cooling, inter-device heat diffusion, and noise.

1.3 Resistance / Hysteresis Model

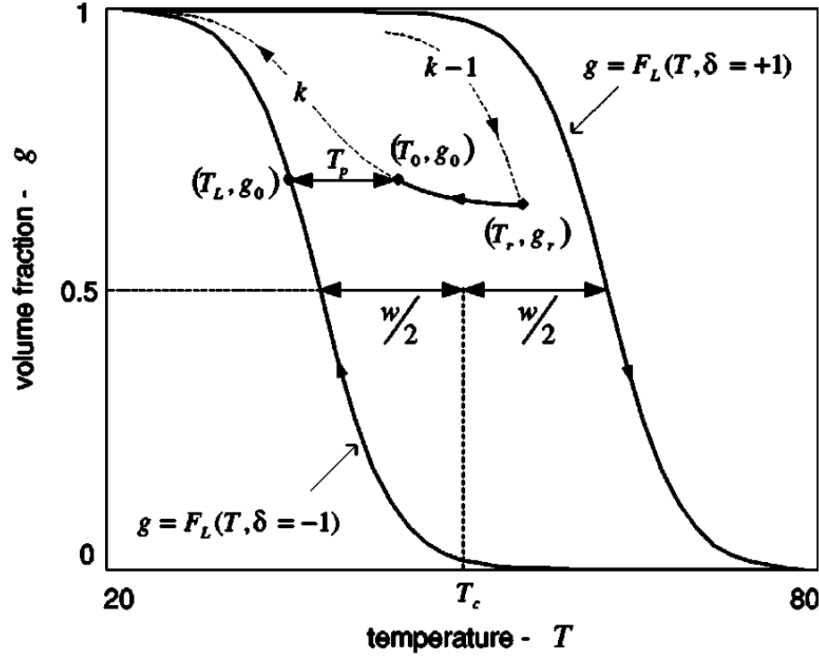


Figure 2: Reference hysteresis geometry showing major branches, reversal points, and proximity temperature shift in the volume-fraction model.

Equation(s)

Starting point from de Almeida *et al.* (2002):

$$R_s(T) = R_0 \exp\left(\frac{E_a}{k_B T}\right). \quad (6)$$

$$R(T) \approx g(T)R_s(T) + (1 - g(T))R_m \Rightarrow R(T) \approx g(T)R_s(T) + R_m. \quad (7)$$

Major-loop semiconducting fraction:

$$g_{\text{major}}(T, \delta) = \frac{1}{2} + \frac{1}{2} \tanh\left[\beta \left(\delta \frac{w}{2} + T_c - T\right)\right], \quad \delta = \begin{cases} +1, & dT/dt > 0, \\ -1, & dT/dt < 0. \end{cases} \quad (8)$$

Minor-loop memory terms:

$$T_{pr} = \delta \frac{w}{2} + T_c - \frac{1}{\beta} \operatorname{arctanh}(2g_r - 1) - T_r, \quad (9)$$

$$x = \frac{T - T_r}{T_{pr}}, \quad P(x) = \frac{1}{2} (1 - \sin(\gamma x)) (1 + \tanh(\pi^2 - 2\pi x)), \quad T_p = T_{pr} P(x), \quad (10)$$

$$g(T, \mathcal{H}) = \frac{1}{2} + \frac{1}{2} \tanh \left[\beta \left(\delta \frac{w}{2} + T_c - (T + T_p) \right) \right]. \quad (11)$$

Final resistance law used in this project:

$$R_{vo2}(T, \mathcal{H}) = R_0 \exp \left(\frac{E_a/k_B}{T} \right) g(T, \mathcal{H}) + R_m. \quad (12)$$

Variables and Units

Symbol	Meaning	Units
$R_{vo2}, R(T)$	VO ₂ device resistance	Ω
$R_s(T)$	semiconducting-branch resistance (Arrhenius)	Ω
R_m	metallic-state resistance floor	Ω
R_0	Arrhenius prefactor	Ω
E_a/k_B	activation-energy-over-Boltzmann term	K
T	device temperature (our model: Kelvin)	K
$g(T, \mathcal{H})$	semiconducting volume fraction with hysteresis memory	–
w	hysteresis width parameter	K
T_c	hysteresis center temperature	K
β	branch steepness	K ^{−1}
δ	branch sign (+1 heating, −1 cooling)	–
T_r, g_r	last reversal state (temperature, fraction)	K, –
T_{pr}	reversal-derived proximity temperature scale	K
γ	proximity-function shape parameter	–
$P(x)$	proximity function controlling minor-loop approach	–
T_p	instantaneous proximity temperature shift	K
$\mathcal{H}$	full hysteresis state ($\delta, T_r, g_r, T_{pr}, \dots$)	–
$\tilde{R}, \tilde{g}$	measured resistance and inferred fraction for fitting	$\Omega, –$

Physical Interpretation

- Eq. (6): $R_s(T)$ decreases exponentially with temperature in the insulating/semiconducting regime.
- Eq. (7): total resistance combines semiconducting contribution gR_s with a metallic floor R_m .
- Eq. (8): two major branches are selected by δ , creating the loop width w .

- Eqs. (9)–(10): reversal variables (T_r, g_r, T_{pr}) and $P(x)$ encode memory and generate minor loops.
- Eq. (12): this is the resistance passed to the electrical/thermal ODEs at every timestep.

Assumptions

- $R_s \gg R_m$ in the transition regime justifies the additive form in Eq. (7).
- Major-loop branches are represented by a symmetric tanh form around T_c .
- Hysteresis state updates only at reversal events (dT/dt sign change).

2 Derivation of the Current-Source Model

2.1 Starting Point: Original Voltage-Driven Equations

Equation(s)

$$C \frac{dV}{dt} = \frac{V_{in}}{R_{series}} - V \left(\frac{1}{R_{vo2}} + \frac{1}{R_{series}} \right), \quad R_{vo2} = R(T, \mathcal{H}). \quad (13)$$

$$C_{th} \frac{dT}{dt} = \frac{V^2}{R_{vo2}} - S_e(T - T_0) + \sigma \eta(t). \quad (14)$$

2.2 Assumptions and Circuit Interpretation

Assumptions

- The voltage source is removed and replaced by a current source in a separate experiment.
- The source enforces an ideal programmed current waveform $I_{in}(t)$.
- An external resistor R_{series} may be present in the injection path in the physical setup.
- Under ideal current drive, adding/removing this external R_{series} does not change node-current balance, so both circuits are dynamically equivalent.
- The VO₂ branch representation is unchanged: $R_{vo2}(T, \mathcal{H})$ in parallel with C , referenced to ground.
- Thermal dynamics keeps the same Joule-heating/cooling structure as in Section 1.2.

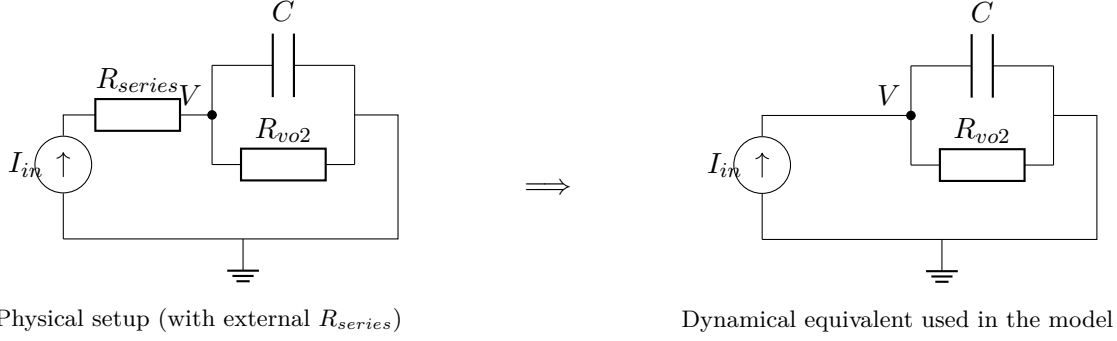


Figure 3: Current-driven setup used in practice (left) and the reduced equivalent used for derivation (right). For ideal enforced $I_{in}(t)$, the external series resistor does not alter node dynamics.

Physical Interpretation

- The imposed source current $I_{in}(t)$ sets the net injected current into node V .
- An external series resistor may drop voltage in the real wiring, but it does not change injected current when the source is ideal.
- At node V , current splits between the capacitor branch and the VO_2 branch.
- VO_2 current V/R_{vo2} remains the electrical-to-thermal coupling path through V^2/R_{vo2} .

2.3 Step-by-Step Derivation for the Direct Current-Source Experiment

Derivation Steps

$$\text{KCL at node } V : \quad I_{in}(t) = I_C + I_{vo2} = C \frac{dV}{dt} + \frac{V}{R_{vo2}}. \quad (15)$$

$$\Rightarrow \quad C \frac{dV}{dt} = I_{in}(t) - \frac{V}{R_{vo2}}. \quad (16)$$

This is identical whether or not an external series resistor is included in the physical current-source lead, provided the source enforces $I_{in}(t)$.

2.4 Final Continuous-Time Current-Driven Model

Equation(s)

$$C \frac{dV}{dt} = I_{in}(t) - \frac{V}{R_{vo2}}, \quad R_{vo2} = R(T, \mathcal{H}). \quad (17)$$

$$C_{th} \frac{dT}{dt} = \frac{V^2}{R_{vo2}} - S_e(T - T_0) + \sigma \eta(t). \quad (18)$$

Variables and Units

Symbol	Meaning	Units
V	node voltage across VO ₂ and capacitor	V
$I_{in}(t)$	injected source current waveform	A
R_{series}	optional external series resistor in the physical setup (non-dynamical here)	Ω
R_{vo2}	hysteretic VO ₂ resistance $R(T, \mathcal{H})$	Ω
C	electrical capacitance at the node	F
T	device temperature	K
C_{th}	thermal capacitance	J/K
S_e	thermal conductance to environment	W/K
T_0	ambient/base temperature	K
σ	thermal-noise amplitude (current module convention)	$\text{W}\sqrt{\text{s}}$
$\eta(t)$	white-noise term	–
$\mathcal{H}$	hysteresis internal state	–

3 Discretization

3.1 Discretization of Current Dynamics Equation

Discrete Update Equation

Using forward Euler with time step Δt , and $R_n = R(T_n, \mathcal{H}_n)$:

$$\left. \frac{dV}{dt} \right|_n = \frac{V_{in,n} - V_n}{R_{series}C} - \frac{V_n}{R_nC}. \quad (19)$$

$$V_{n+1} = V_n + \Delta t \left. \frac{dV}{dt} \right|_n. \quad (20)$$

Equivalent compact form:

$$V_{n+1} = V_n + \frac{\Delta t}{C} \left[\frac{V_{in,n}}{R_{series}} - V_n \left(\frac{1}{R_n} + \frac{1}{R_{series}} \right) \right]. \quad (21)$$

State Update Order

1. Evaluate hysteretic resistance R_n from current thermal state T_n and hysteresis memory $\mathcal{H}_n$.
2. Compute dV/dt at step n from Eq. (19).
3. Update node voltage to V_{n+1} (explicit Euler).

3.2 Discretization of Thermal Dynamics Equation

Discrete Update Equation

Define $P_n = V_n^2/R_n$. The thermal derivative at step n is:

$$\left. \frac{dT}{dt} \right|_n = \frac{P_n - S_{env}(T_n - T_0) + S_{couple} \nabla_d^2 T_n}{C_{th}} + \xi_n. \quad (22)$$

Forward Euler update:

$$T_{n+1} = T_n + \Delta t \left. \frac{dT}{dt} \right|_n. \quad (23)$$

where ∇_d^2 is the discrete Laplacian and ξ_n is optional additive temperature-rate noise.

For the simplified single-device non-deterministic case ($S_{couple} = 0$):

$$\left. \frac{dT}{dt} \right|_n = \frac{1}{C_{th}} \left(\frac{V_n^2}{R_n} - S_e(T_n - T_0) \right) + \xi_n, \quad (24)$$

$$T_{n+1} = T_n + \Delta t \left. \frac{dT}{dt} \right|_n. \quad (25)$$

State Update Order

1. Use electrical state V_n and resistance R_n to compute Joule power $P_n = V_n^2/R_n$.
2. Compute thermal derivative from Eq. (22) (or Eq. (24) in the reduced case).
3. Update temperature to T_{n+1} (explicit Euler).

3.3 Discretization of Resistance / Hysteresis Model

Discrete State Definitions

$$\mathcal{H}_n = \left\{ \delta_n, T_{r,n}, g_{r,n}, T_{pr,n}, \text{reversed}_n, T_n^{\text{last}} \right\}. \quad (26)$$

Given T_n and $\mathcal{H}_n$, the discrete evaluation is:

$$T_{p,n} = T_{pr,n} P \left(\frac{T_n - T_{r,n}}{T_{pr,n}} \right) \text{reversed}_n, \quad (27)$$

In implementation, a small ε regularization is used in the denominator when $|T_{pr,n}|$ is very small.

$$g_n = \frac{1}{2} + \frac{1}{2} \tanh \left[\beta \left(\delta_n \frac{w}{2} + T_c - (T_n + T_{p,n}) \right) \right], \quad (28)$$

$$R_n = R_0 \exp \left(\frac{E_a/k_B}{T_n} \right) g_n + R_m. \quad (29)$$

Update Order (including reversals)

1. Compute $\Delta T_n = T_n - T_n^{last}$. If $|\Delta T_n|$ exceeds threshold and $\text{sign}(\Delta T_n) \neq \delta_n$, trigger reversal.

2. At reversal, set:

$$\delta_n \leftarrow \text{sign}(\Delta T_n), \quad g_{r,n} \leftarrow g(T_n, \mathcal{H}_n^-), \quad T_{r,n} \leftarrow T_n, \quad (30)$$

$$T_{pr,n} \leftarrow \delta_n \frac{w}{2} + T_c - \frac{1}{\beta} \text{arctanh}(2g_{r,n} - 1) - T_{r,n}, \quad \text{reversed}_n \leftarrow 1. \quad (31)$$

3. Evaluate g_n and R_n from Eqs. (28)–(29).

4. Set $T_{n+1}^{last} \leftarrow T_n$ for the next reversal check.

3.4 Discretization of the Current-Source Model

Discrete Update Equation

For the direct ideal-current model of Section 2:

$$V_{n+1} = V_n + \frac{\Delta t}{C} \left(I_{in,n} - \frac{V_n}{R_n} \right), \quad R_n = R(T_n, \mathcal{H}_n). \quad (32)$$

Thermal update in the current module (Euler-Maruyama in implementation):

$$T_{n+1} = T_n + \frac{\Delta t}{C_{th}} \left(\frac{V_n^2}{R_n} - S_e(T_n - T_0) \right) + \frac{\sigma}{C_{th}} \sqrt{\Delta t} \zeta_n, \quad \zeta_n \sim \mathcal{N}(0, 1). \quad (33)$$

State Update Order

1. Evaluate R_n from $(T_n, \mathcal{H}_n)$.
2. Update V_{n+1} with Eq. (32).
3. Update T_{n+1} with Eq. (33).
4. Update hysteresis memory using (T_n, T_{n+1}) for the next step.

4 Fitting of the Resistance Model Based on Real Data

4.1 Input Data and Preprocessing

What is directly measured

The raw export provides eight columns per sample:

$$\{T, R, t, I, V_+, V_-, T_+, T_-\}$$

with units $\{\text{K}, \Omega, \text{s}, \text{A}, \text{V}, \text{V}, \text{K}, \text{K}\}$.

Table 1: Representative rows from the experimental dataset (actual values).

T (K)	R (Ω)	t (s)	I (A)	V_+ (V)	V_- (V)	T_+ (K)	T_- (K)
369.8230	17.6701	161.6042	2.0×10^{-6}	5.42×10^{-5}	-1.65×10^{-5}	369.818	369.828
338.0850	30.9270	399.6742	2.0×10^{-6}	6.23×10^{-5}	-6.14×10^{-5}	338.212	337.958
305.7635	3288.1260	642.2995	2.0×10^{-6}	6.55×10^{-3}	-6.61×10^{-3}	305.892	305.635
289.8545	4625.3550	824.9370	2.0×10^{-6}	9.23×10^{-3}	-9.27×10^{-3}	289.845	289.864
305.2965	3485.7080	939.4225	2.0×10^{-6}	6.98×10^{-3}	-6.96×10^{-3}	305.168	305.425
370.1625	17.5121	1424.8130	2.0×10^{-6}	5.63×10^{-5}	-1.38×10^{-5}	370.036	370.289

What is cleaned/derived

Preprocessing is:

1. Remove the units row (the row containing K, Ohm, sec, ...).
2. Convert all columns to numeric and drop invalid rows.
3. Keep rows with valid Temperature and Resistance; sort by Time.
4. If $\text{median}(T) < 200$, interpret T as $^{\circ}\text{C}$ and convert to K.

Then we compute directional masks from

$$\Delta T_n = T_n - T_{n-1}, \quad \text{cooling if } \Delta T_n < 0, \quad \text{heating if } \Delta T_n > 0. \quad (34)$$

For this dataset after cleaning:

$$N = 264, \quad T \in [289.8545, 370.1625] \text{ K}, \quad R \in [17.3765, 4625.355] \Omega.$$

4.2 What We Obtain at Each Step vs What Must Be Assumed

Table 2: Given vs computed vs unknown quantities in the resistance fit pipeline.

Class	Quantities
Given by experiment	$T(t)$, $R(t)$, time stamps, and auxiliary channels (I , V_+ , V_- , T_+ , T_-).
Computed directly from data	ΔT_n , cooling/heating masks, sweep turning point, and dataset bounds ($T_{\min}$, $T_{\max}$, $R_{\min}$, $R_{\max}$).
Unknown after preprocessing (must be fitted or swept)	R_0 , E_a/k_B , R_m , w , T_c , β , start branch (insulator or metal), and optionally γ if enough minor-loop information exists.

4.3 Parameter Seeding and Intermediate Quantities

Step-by-step outputs

The implemented seeding path is:

1. **Seed** ($R_0, E_a/k_B, R_m$) **from extremes.** A robust low-temperature window is selected; then R_m is scanned and $\ln(R - R_m)$ is linearly regressed versus $1/T$. For this dataset:

$$R_0^{seed} = 10.9606, \quad (E_a/k_B)^{seed} = 1753.57 \text{ K}, \quad R_m^{seed} = 0.8688 \Omega.$$

2. **Build seed semiconducting fraction.**

$$g_n^{seed} = \text{clip}\left(\frac{R_n - R_m^{seed}}{R_0^{seed} \exp((E_a/k_B)^{seed}/T_n)}, 0, 1\right). \quad (35)$$

3. **Seed major-loop geometry** (w, T_c). Cooling/heating temperatures closest to $g^{seed} = 0.5$ are used:

$$w^{seed} = 5.804 \text{ K}, \quad T_c^{seed} = 331.235 \text{ K}.$$

4. **Seed start branch.** From the first non-zero ΔT : `start_branch_hint = metal`.

4.4 Optimization Objective and Fitting Procedure

Objective function

The optimizer minimizes, for parameter vector x and branch b :

$$\mathcal{L}(x, b) = \text{RMSE}_{\log R}(x, b) + \lambda_g \text{RMSE}_g(x, b) + \mathcal{P}_\gamma, \quad \lambda_g = 0.2. \quad (36)$$

with

$$\text{RMSE}_{\log R} = \sqrt{\frac{1}{N} \sum_{n=1}^N [\log_{10} R_n^{model}(x, b) - \log_{10} R_n^{data}]^2}, \quad (37)$$

$$\text{RMSE}_g = \sqrt{\frac{1}{N} \sum_{n=1}^N [g_n^{model}(x, b) - g_n^{exp}(x)]^2}, \quad (38)$$

$$g_n^{exp}(x) = \text{clip}\left(\frac{R_n^{data} - R_m(x)}{R_0(x) \exp((E_a/k_B)(x)/T_n)}, 0, 1\right). \quad (39)$$

When `--fit-gamma` is active:

$$\mathcal{P}_\gamma = 0.01 |\gamma - \gamma_{\text{fixed}}|,$$

otherwise $\mathcal{P}_\gamma = 0$.

Meaning of Terms and Acronyms

- **RMSE:** *Root Mean Squared Error*. It is the square root of the mean squared residual.
- $\text{RMSE}_{\log R}$: RMSE computed on $\log_{10}(R)$, so multiplicative resistance mismatches are penalized more uniformly across decades.

- RMSE_g : RMSE between model and reconstructed semiconducting fraction g , used to keep hysteresis-shape consistency.
- λ_g : scalar weight that controls the influence of the g -consistency term relative to resistance fitting ($\lambda_g = 0.2$ in this work).
- $\mathcal{P}_\gamma$: optional regularization term for γ , used only when `--fit-gamma` is enabled.
- $\text{clip}(u, 0, 1)$: saturation/projection operator to enforce physical bounds for a fraction:

$$\text{clip}(u, 0, 1) = \begin{cases} 0, & u < 0, \\ u, & 0 \leq u \leq 1, \\ 1, & u > 1. \end{cases}$$

- x : optimization vector containing fitted parameters (e.g., $R_0, E_a/k_B, R_m, w, T_c, \beta$, optionally γ).
- b : initial hysteresis branch choice (`insulator` or `metal`) used when evaluating a candidate x .

This optimizer is a custom implementation. Methodologically, it is a bounded derivative-free search (random exploration + coordinate local refinement), conceptually related to direct-search/pattern-search methods (see Sources 3–5) and standard nonlinear optimization practice (Source 6).

Algorithm

1. **Bounded initialization:** define lower/upper bounds for each component of x , then build initial candidates from (i) baseline values and (ii) data-driven seeds.
2. **Branch screening:** evaluate candidates on both initial branches (`insulator`, `metal`) and keep the currently best pair (x, b) .
3. **Global exploration (random search):** for many iterations, propose new points by either: (a) Gaussian perturbation around current best, or (b) uniform draw in the full box. Every proposal is projected back into bounds componentwise by clipping.
4. **Local refinement (coordinate search):** update one coordinate at a time with $\pm$ step moves; accept only improving moves. If no coordinate improves, shrink step size and continue.
5. **Final branch check:** with the best x , evaluate both branches once more and keep the better one.
6. **Report:** output final parameters and RMSE metrics (overall, cooling-only, heating-only).

4.5 Final Fitted Parameters, Quality Metrics, and Limits

Results

For the fitted case documented here:

Table 3: Fitted resistance/hysteresis parameters.

Parameter	Value
R_0	0.6461899358
E_a/k_B (K)	2606.0865549
R_m (Ω)	18.3025411908
w (K)	6.9842869901
T_c (K)	333.4693743794
β	0.2915514718
γ	0.9562696820 (fixed in this run)
Start branch	metal

Table 4: Fit quality metrics ($\log_{10} R$ RMSE).

Metric	Value
Overall RMSE	0.0373334
Cooling RMSE	0.0360230
Heating RMSE	0.0387234

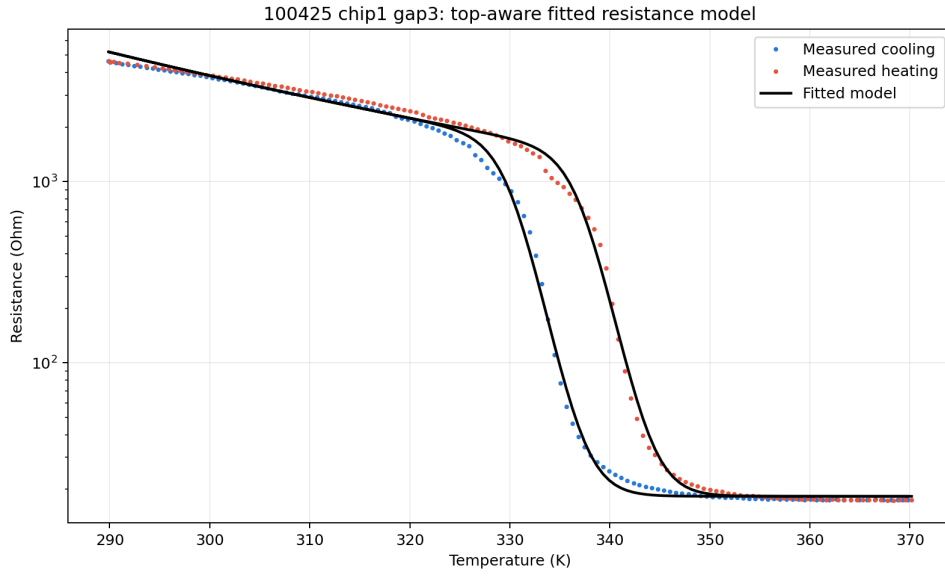


Figure 4: Measured $R(T)$ points and fitted model for the dataset used in Section 4.

Limitations

At the end of this procedure, the following still require additional identification work:

- **Minor-loop fidelity:** one major cooling-heating cycle does not fully constrain history parameters; dedicated reversal experiments are needed.
- **γ identifiability:** with major-loop data only, γ is weakly identifiable, so it is fixed unless additional data is provided.
- **Parameter uniqueness:** different parameter tuples can produce very similar $\log R(T)$ errors; confidence intervals are not yet computed.
- **Not part of this fit:** electrical/thermal dynamical parameters (C , C_{th} , S_e , source terms) must be calibrated separately from transient simulations.

5 Quasistatic and Dynamic-Phase Current-Source Models

5.1 Shared electrothermal equations

The ideal current-source circuit uses

$$C \frac{dV}{dt} = I_{\text{in}} - \frac{V}{R}, \quad (40)$$

$$C_{\text{th}} \frac{dT}{dt} = \frac{V^2}{R} - S_e(T - T_0). \quad (41)$$

Both the quasistatic and dynamic-phase models use these same electrical and thermal balances and the same fitted resistance law

$$R(T, g) = R_m + R_0 \exp\left(\frac{E_a/k_B}{T}\right) g. \quad (42)$$

The only distinction is how the actual insulating phase fraction g responds to the hysteresis-prescribed equilibrium fraction $g_{\text{eq}}(T, \mathcal{H})$.

5.2 Original quasistatic phase response

The original Yuanhang Zhang model assumes that the phase fraction responds immediately:

$$g(t) = g_{\text{eq}}(T(t), \mathcal{H}(t)). \quad (43)$$

The independent dynamical states are therefore V and T . Whether a numerical trajectory follows the intended minor-loop geometry also depends on faithful implementation of the discrete reversal detector, as documented below.

A matched quasistatic numerical control also demonstrated why visual inspection of one trace is insufficient. At a reference timestep, several currents showed repeated switching that looked oscillatory. However, changing only the integration timestep caused those cycles to appear, disappear, or change sharply in frequency and amplitude. For example, the 1100 μA control showed

repeated cycles at $dt = 0.025$ ns, no cycles at $dt = 0.05$ ns, and a different large switching event at finer timesteps. These are apparent numerical oscillations, not a timestep-converged physical limit cycle.

The application History entry `Quasistatic apparent oscillations - numerical control` preserves these traces with the standard current slider. They were produced by the historical local port bug, not by the exact upstream reversal algorithm.

5.3 Dynamic-phase working extension

The local dynamic-phase extension treats g as a third dynamical state:

$$\tau_g \frac{dg}{dt} = g_{\text{eq}}(T, \mathcal{H}) - g. \quad (44)$$

The resistance in Equation 42 is evaluated using this delayed $g(t)$. The quasistatic model is recovered as $\tau_g \rightarrow 0$.

Practically, finite τ_g means that temperature first changes the preferred phase fraction, while the actual material state and resistance respond later. This introduces a delayed feedback loop:

$$I_{\text{in}} \rightarrow V \rightarrow P_{\text{Joule}} \rightarrow T \rightarrow g_{\text{eq}} \rightarrow g \rightarrow R \rightarrow V.$$

The delay can support a limit cycle even when the corresponding instantaneous-feedback model approaches a fixed point. This extension is physically motivated by finite transition kinetics, but it is not part of the original Yuanhang Zhang repository and must be independently calibrated before its quantitative predictions are treated as experimental predictions.

Table 5: Direct comparison of the two phase-response assumptions.

Property	Quasistatic model	Dynamic-phase extension
Independent states	V, T	V, T, g
Phase response	$g = g_{\text{eq}}(T, \mathcal{H})$	$\tau_g \dot{g} = g_{\text{eq}} - g$
New parameter	none	τ_g
Original Yuanhang code	yes	no
Current-source result after reversal-detector fix	a newly targeted parameter set supports a converged relaxation-oscillation domain	tested prior candidate points approach fixed points

6 Dynamic-Model Article-Comparison Campaign

6.1 Target observables

The comparison preprint supplied for this project reports a current-driven oscillatory regime at approximately 200–600 μA , oscillation frequencies near 40–60 MHz, mean power near 60–100 μW , and energy near 1.2–1.9 pJ per cycle. These values were used as targets, not as fitted constraints.

6.2 Search procedure

The search kept the resistance model and dynamic-phase equations unchanged. It:

1. used the improved measured-sample resistance fits;
2. broadly sampled C , C_{th} , S_e , τ_g , and T_0 ;
3. first screened the center of the reported current window;
4. tested neighboring currents only after a candidate showed repeated cycles; and
5. required late-window persistence and repeated simulations at smaller timesteps before accepting a result.

Across the four newly fitted samples and the best-characterized 100425 chip1 gap3 sample, the broad 200–600 μA searches produced large switching transients but no sustained 40–60 MHz limit cycle.

At a higher-current exploratory base point,

$$C = 80 \text{ pF}, \quad C_{\text{th}} = 3 \text{ pJ/K}, \quad S_e = 0.052838 \text{ mW/K}, \quad \tau_g = 10 \text{ ns}, \quad T_0 = 325 \text{ K},$$

coarser integrations using the historical local per-step port showed apparent late-time oscillations over parts of approximately 0.8–1.8 mA, with frequencies of roughly 4.7–10.1 MHz. These legacy results did not quantitatively reproduce the preprint’s current or frequency range and did not survive corrected reversal handling.

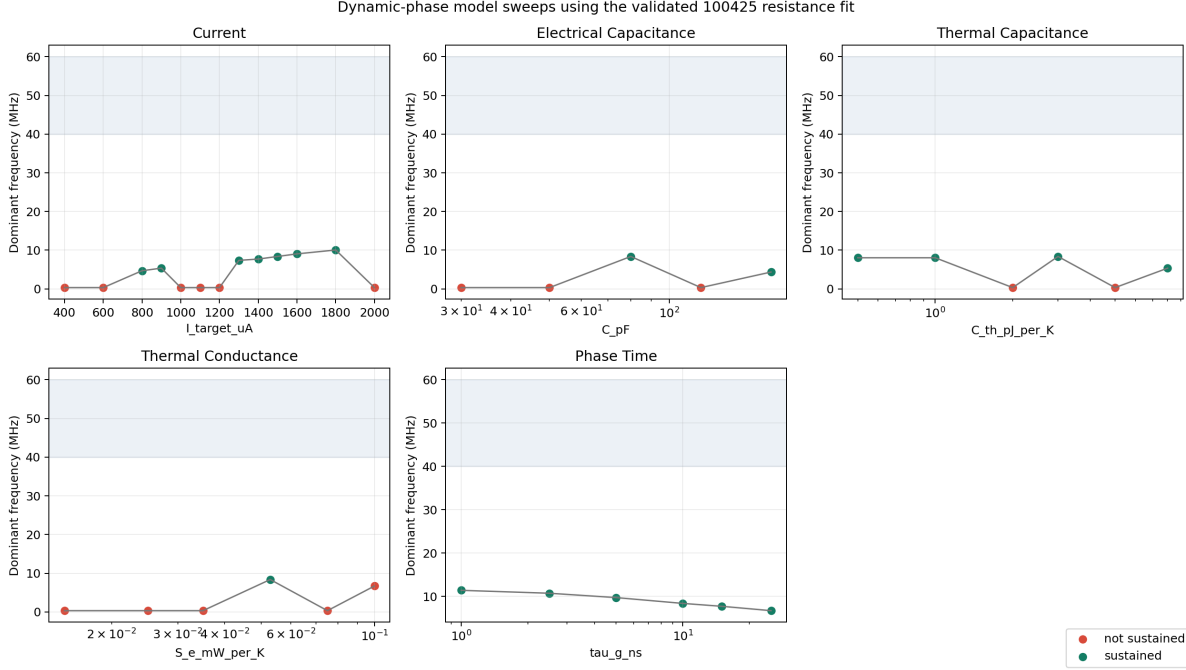


Figure 5: Legacy one-at-a-time dynamic-phase sweeps generated with the timestep-dependent local port bug. Green points passed the late-window criterion at the screening timestep, but the apparent oscillations did not survive corrected reversal handling.

6.3 Timestep-convergence audit and numerical correction

The apparent oscillatory domains were tested over $5 \mu\text{s}$ at timesteps of 0.05, 0.025, and 0.0125 ns. The 0.05 and 0.025 ns results were often close, but the 0.0125 ns results changed sharply in frequency and voltage amplitude. Therefore, the apparent domains are not yet timestep-converged.

The sensitivity was associated with a bug in the local port of the hysteresis reversal detector. The port updated its temperature anchor after every numerical step, so a reversal was accepted only when the temperature change in one integration step exceeded the fixed threshold

$$|\Delta T_{\text{step}}| > 0.01 \text{ K}. \quad (45)$$

Changing the timestep changes ΔT_{step} , and therefore changed whether and when the hysteresis branch reversed. This was not faithful to Yuanhang Zhang's source code. Upstream updates its anchor only after accumulated displacement exceeds 0.01 K.

Two corrected modes are now explicit. The default **yuanhang_reference** mode faithfully implements upstream accumulated-displacement and sampled detection-point geometry. The optional **turning_point** extension tracks the latest temperature extremum T_{extreme} and confirms a reversal after the cumulative excursion satisfies

$$|T - T_{\text{extreme}}| > 0.01 \text{ K}. \quad (46)$$

In the extension, the reversal point is assigned to T_{extreme} , rather than to the sampled detection point used upstream. This makes the extension timestep robust but changes minor-loop geometry. A third mode, `ported_per_step`, is retained only to reproduce historical artifacts and must not be used for new physical claims.

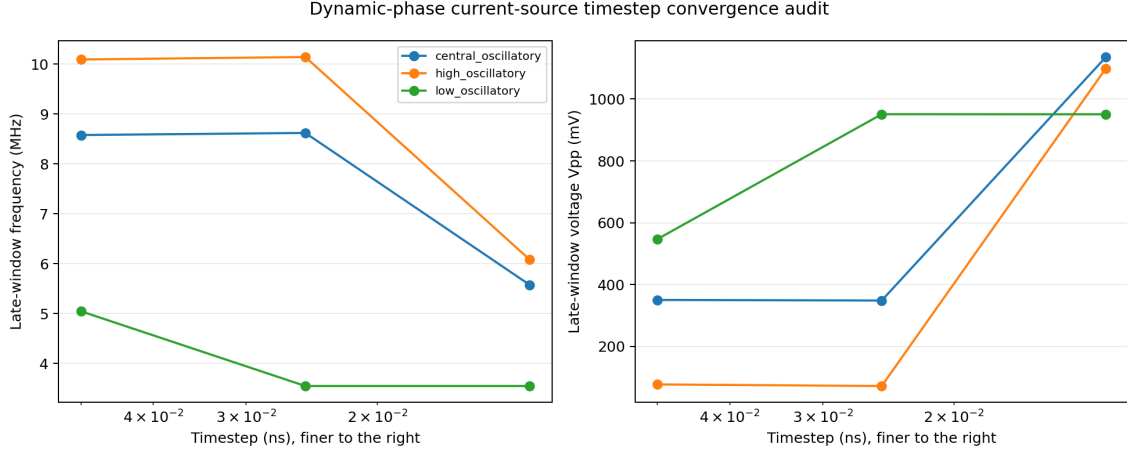


Figure 6: Legacy timestep-convergence audit for representative apparent oscillatory points. The frequency and amplitude changes exposed the local anchor-update bug and motivated a full upstream-fidelity audit.

6.4 Post-correction convergence results

The measured 100425 resistance-fit RMSE remained effectively unchanged at 0.03733344, showing that the numerical correction did not disturb the fitted major resistance loop.

After the correction, the previously tested quasistatic and dynamic-phase current-source candidate points at 800, 1100, 1500, and 1800 μA all approached fixed points consistently for $dt = 0.05, 0.025, 0.0125,$ and 0.00625 ns. The previously reported domains at those parameter sets must therefore be classified as numerical artifacts of the legacy reversal detector.

A subsequent first-principles search using the `turning_point` reversal extension found a different, timestep-converged oscillatory domain while retaining the original quasistatic ideal-current equations and fixed measured resistance fit. The search targeted the steep negative-slope portion of the fitted heating branch. At a current-source equilibrium,

$$I^2 R(T) = S_e (T - T_0), \quad (47)$$

and the local V, T Jacobian can become an unstable focus when the negative dR/dT feedback overcomes electrical and thermal damping. Simulating parameter sets selected from this condition identified

$$C = 25.930953 \text{ pF}, \quad C_{\text{th}} = 5.0 \text{ pJ/K}, \quad S_e = 0.1 \text{ mW/K}, \quad T_0 = 311.219374 \text{ K}.$$

For the measured 100425 chip1 gap3 resistance fit, that extension produced sustained quasistatic relaxation oscillations from approximately 1.27 to 2.74 mA. Fixed points were recovered at 1.20 and 2.80 mA. Across the extension’s oscillatory domain, frequency increased from approximately 5.49 to 14.13 MHz. A subsequent exact-reference resweep found a narrower upper edge: sustained late-window oscillations remained at 2.47 mA, while 2.48 mA settled to a fixed point.

At $I = 1640.226 \mu\text{A}$, the period converges as the timestep is halved:

dt (ns)	Period (ns)	Frequency (MHz)
0.015820	123.206	8.116
0.007910	123.135	8.121
0.003955	123.095	8.124

The late-window voltage peak-to-peak amplitude converges near 2.275 V, and the temperature remains approximately 325.4–349.1 K. The application History entry **Validated old-model quasistatic oscillatory domain** preserves fixed-point controls and representative traces, but its name pre-dates the fidelity audit: those saved traces used the **turning_point** extension.

As a positive control, the original voltage-driven relaxation oscillator remained oscillatory and converged as the timestep was reduced:

dt (ns)	Frequency (MHz)	Voltage peak-to-peak (V)
10.00	0.4330	5.012
5.00	0.4372	4.921
2.50	0.4395	4.873
1.25	0.4407	4.849

This is the expected signature of a robust circuit-driven oscillation: the behavior persists and its quantitative values approach stable limits as dt decreases.

6.5 Current scientific conclusion

The quasistatic ideal-current equations support a broad, timestep-converged relaxation-oscillation domain when the electrical and thermal timescales are selected from the instability condition on the measured resistance transition. The sole supported hysteresis implementation is now the exact Yuanhang reference detector, for which the audited sweep extends through approximately 2.47 mA and reaches a fixed point at 2.48 mA. The former turning-point extension produced an historical upper boundary near 2.74 mA, but it was removed from executable code because it changed minor-loop geometry. The dynamic-phase extension remains a physically plausible hypothesis, but the searched dynamic-phase parameter sets have not yet produced a separately validated current-source limit cycle. Specifically:

- the searched model did not reproduce the supplied preprint’s 200–600 μA , 40–60 MHz domain;

- the old candidate domains disappeared after correcting timestep-dependent reversal recognition;
- an historical turning-point search found a different domain, but that reversal extension is no longer executable;
- the exact-reference resweep found oscillations through 2.47 mA and a fixed point at 2.48 mA;
- the exact upstream float32 hysteresis path is now the only software implementation and matches the vendored reference trace to within 0.001 Ω on the fidelity audit;
- the original voltage-driven oscillator survives and converges under the same correction; and
- all future positive domains must continue to pass corrected-detector timestep and duration checks.

Legacy exploratory traces remain saved in the application History and are explicitly labeled as pre-correction numerical results. Their obsolete mode fields are provenance only; rerunning a historical configuration uses the sole Yuanhang reference implementation.

7 Sources

1. Y.-H. Zhang *et al.*, “Collective dynamics and long-range order in thermal neuristor networks,” *Nature Communications*, 2024.
2. L. A. L. de Almeida *et al.*, “Modeling of the hysteretic metal-insulator transition in a vanadium dioxide infrared detector,” *Optical Engineering*, 2002.
3. R. Hooke and T. A. Jeeves, “Direct Search” Solution of Numerical and Statistical Problems,” *Journal of the ACM*, vol. 8, no. 2, pp. 212–229, 1961.
4. T. G. Kolda, R. M. Lewis, and V. Torczon, “Optimization by Direct Search: New Perspectives on Some Classical and Modern Methods,” *SIAM Review*, vol. 45, no. 3, pp. 385–482, 2003.
5. A. R. Conn, K. Scheinberg, and L. N. Vicente, *Introduction to Derivative-Free Optimization*. SIAM, 2009.
6. “Harnessing the VO₂ Phase Transition for Automatic Gain Control in Transimpedance Amplifiers,” preprint arXiv:2604.04594, 2026.